

Basanta Khakurel, M.S.

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EDUCATION

Ph.D. Earth and Environmental Sciences

Ludwig-Maximilians-Universität München

Munich, Germany

Aug 2026 (Expected)

Supervisor: Dr. Sebastian Höhna

Thesis: Toward More Realistic Models of Morphological Evolution in Bayesian Phylogenetics.

M.S. Biology

Southeastern Louisiana University

Hammond, LA, US

May 2023

GPA: 4.0

Supervisor: Dr. April Wright

Thesis: Inference of Phylogeny using Discrete Morphological Characters.

B.S. Biology (Minors: Chemistry & Political Science)

Southeastern Louisiana University

Hammond, LA, US

May 2021

Advisor: Dr. April Wright

*Thesis: Species delimitation of Eastern Pinesnakes (*Pituophis melanoleucus*).*

RESEARCH EXPERIENCE

Research Assistant

Höhna Lab, Department of Earth and Environmental Sciences,
Ludwig-Maximilians-Universität München

Munich, Germany

Sep 2023 – Present

- Bayesian phylogenetic methods for morphological datasets

Graduate Research Assistant

Wright Lab, Department of Biological Sciences,
Southeastern Louisiana University

Hammond, LA, US

Jun 2021 – May 2023

- Utilized RevBayes and R for exploring details on morphological models for phylogenetic inference.
- Posterior predictive simulations in phylogenetics.

Undergraduate Research Assistant

Wright Lab, Department of Biological Sciences,
Southeastern Louisiana University

Hammond, LA, US

Aug 2019 – May 2021

- Performed species delimitation analyses on eastern Pinesnake.
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TECHNICAL COMPETENCIES

Coding:

R, RevBayes, C++, Rust, LaTeX, Typst, sh, HTML/CSS, Git.

Methods:

Bayesian Inference, Phylogenetics, Morphological Modeling, Fossil Tip-Dating, Posterior Predictive Simulations, Species Delimitation, Inkscape Graphics.

Languages:

Nepali (Native), English (Fluent), Hindi (Fluent), German (Basic)

RESEARCH PUBLICATIONS

Submitted Manuscripts

1. **Khakurel B.**, Capobianco A., Höhna S. “The hidden cost of discretization: How the number of rate categories affect phylogenetic inference.” *Royal Society Open Science* (Submitted). 2026
 2. **Khakurel B.**, Darlim G., Capobianco A., Höhna S. “Character level insights on the mosaic nature of morphological evolution.” *Proceedings of Royal Society B* (Submitted). 2026
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Journal Articles

1. Wynd B.M., **Khakurel B.**, Kammerer C.F., Wagner P.J., Wright A.M. “Incorporating continuous characters in joint estimation of dicynodont phylogeny.” *Systematic Biology*. 2026
<https://doi.org/10.1093/sysbio/syaf078>
2. **Khakurel B.**, Darlim G., Capobianco A., Höhna S. “A covarion model for phylogenetic estimation using discrete morphological datasets.” *Systematic Biology*. 2026
<https://doi.org/10.1093/sysbio/syag014>
3. Heckeberg N., Capobianco A., **Khakurel B.**, Darlim G., Höhna S. “Practical Guide and Review of Fossil Tip-Dating in Phylogenetics.” *Systematic Biology*. 2026
<https://doi.org/10.1093/sysbio/syaf050>
4. Girón Duque JC, Balk M, Dahdul W, Lapp H, Mikó I, Alhajjar E, Wynd B, Tarasov S, Lawrence C, **Khakurel B.**, Porto A, Yan L, Fluck IE, Porto D, Keating J, Borokini I, Seltmann K, Montanaro G, Mabee P. “Meeting Report for the Phenoscape TraitFest 2023 with Comments on Organising Interdisciplinary Meetings.” *Biodiversity Information Science and Standards*. 2024
<https://doi.org/10.3897/biss.8.115232>

5. **Khakurel B.**, Grigsby C., Tran T.D., Zariwala J., Wright A.M. “The fundamental role of character coding in Bayesian morphological phylogenetics.” *Systematic Biology*. 2024
<https://doi.org/10.1093/sysbio/syae033>
 6. **Khakurel B.**, Nikolakis Z.L., Crother B.I., Wright A.M. “Species delimitation of eastern Pinesnake complex (*Pituophis melanoleucus*).” *Bulletin of the Society of Systematic Biologists*. 2023
<https://doi.org/10.18061/bssb.v2i1.9423>
 7. Ward S.J., McMahan C.D., **Khakurel B.**, Wright A.M, Piller KR. “Genomic data support the taxonomic validity of Middle American livebearers *Poeciliopsis gracilis* and *Poeciliopsis pleurospilus* (Cyprinodontiformes: Poeciliidae).” *PloS one*. 2022
<https://doi.org/10.1371/journal.pone.0262687>
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TEACHING & MENTORING

Invited Lecturer

LMU München

Munich, Germany

Jan 2026

- Phylogenetic Analysis of Molecules and Morphology (Modeling Morphological Evolution).

Invited Lecturer

Southeastern Louisiana University

Hammond, LA, US

Sep 2023

- Biological Data Analysis Class (Simulations in Biology).

Summer Camp Teacher

Upward Bound Math and Science Program

Hammond, LA, US

Jun 2019

- Taught mathematics and biology to high school students.
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PRESENTATION AND TALKS

Invited Talks

1. **B. Khakurel**, and Sebastian Höhna, “A Covarion model for Discrete Morphological Datasets, Beyond the Mk model: Addressing Heterotachy,” in *Bristol Paleobiology Group*, University of Bristol, Jun. 2026.
2. **B. Khakurel**, “Snakes, Fish, and Bayesian morphological phylogenetics,” in *PhD and PostDoc Seminar*, Paleontological Museum Munich, Jul. 2024.

3. **B. Khakurel**, B. M. Wynd, and A. M. Wright, "Use of Continuous Traits in Phylogenetic Estimation," in *Departmental Seminar*, Department of Biological Sciences, Southeastern Louisiana University, Apr. 2023.
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Presentations at Academic Conferences

1. **B. Khakurel**, A. Capobianco, and S. Höhna, "The impact of rate variation assumptions on phylogenetic accuracy," in *EvoGen Munich*, Munich, Germany, Mar. 2026.
2. **B. Khakurel**, A. Capobianco, and S. Höhna, "The Hidden Cost of Discretization: Is the default $k=4$ biasing your tree lengths?" In *Society of Systematic Biologists Meeting*, Baton Rouge, LA, USA, Jan. 2026.
3. **B. Khakurel** and S. Höhna, "Characters changing speed: the Covariomorph model and its impact on phylogenetic trees," in *ESEB 2025 Congress*, Poster presentation, Barcelona, Spain, Aug. 2025.
4. **B. Khakurel** and S. Höhna, "Characters changing speed: the Covariomorph model and its impact on phylogenetic trees," in *Evolution Meetings*, Athens, Georgia, USA, Jun. 2025.
5. **B. Khakurel** and S. Höhna, "An Efficient Heterotachy Model for Morphological Phylogenetics," in *3rd Joint Congress on Evolutionary Biology*, Montréal, Canada, Jul. 2024.
6. **B. Khakurel** and S. Höhna, "Evidence of covarion rate variation in empirical morphological datasets," in *The Paleontological Association Annual Meeting*, Poster presentation, Erlangen, Germany, Dec. 2024.
7. N. Heckeberg, A. Capobianco, G. Darlim, **B. Khakurel**, and S. Höhna, "Potential and Perils in Using Morphological Characters for Phylogenetic Inference," in *The Palaeontological Association Annual Meeting*, Poster presentation, Cambridge, UK, 2023.
8. **B. Khakurel** and A. M. Wright, "Model Adequacy in Morphological Data," in *Society of Systematic Biologists Meeting*, Mexico City, Mexico, Jan. 2023.
9. **B. Khakurel**, B. M. Wynd, and A. M. Wright, "Use of Continuous Traits in Phylogenetic Estimation," in *Evolution Meetings*, Albuquerque, NM, USA, Jun. 2023.
10. B. M. Wynd, **B. Khakurel**, and A. M. Wright, "Incorporating Continuous Characters in Total-Evidence Phylogenetic Estimation of Extinct Lineages," in *GSA Connects Meeting*, Pittsburgh, PA, USA, 2023.
11. **B. Khakurel** and A. M. Wright, "Species Delimitation in Eastern Pine Snakes (*Pituophis melanoleucus*)," in *Evolution Meetings*, Cleveland, OH, USA, Jun. 2022.
12. O. Schwery, **B. Khakurel**, and A. M. Wright, "Adequacy of Fossilised Birth-Death Models: A Test Case in Ants," in *Evolution Meetings*, Cleveland, OH, USA, Jun. 2022.
13. **B. Khakurel** and A. M. Wright, "Species Delimitation in Eastern Pine Snakes (*Pituophis melanoleucus*)," in *LBRN 8th Computational Biology and Bioinformatics Meeting*, Baton Rouge, LA, USA, 2021.

14. **B. Khakurel**, C. Grigsby, T. Tran, and A. M. Wright, "Character State Space Partitioning in Phylogenetic Analyses from Discrete Data," in *Louisiana Biomedical Research Network 18th Annual Meeting*, Poster presentation, Baton Rouge, LA, USA, 2020.
 15. **B. Khakurel**, C. Grigsby, and A. M. Wright, "Modeling Morphological State Space," in *Society of Systematic Biologists Meeting*, Poster presentation, Gainesville, FL, USA, Jan. 2020.
 16. **B. Khakurel**, V. Rocquin, and J. Litloff, "The Role of Honors Student Organizations in Building and Sustaining a Robust Honors Presence on Campus," in *Southern Regional Honors Council Conference*, Memphis, TN, USA, 2019.
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SERVICE, OUTREACH & ADVANCED TRAINING

Advanced Training & Computational Workshops

- A Primer for Phylogenetic Causal Inference Workshop
 - Generalized Bayesian Phylogenetics to Infer Simultaneous and Multifurcating Divergences
 - Phylogenetic methods for Historical Biogeography
 - Visualizing Science Workshop Series
 - Navigating Scientific Publishing - Early Career Focus
 - Phenoscape Traitfest
 - RevBayes Workshop and Hackathon
 - PhylogatR (Phylogeographic data Aggregation and Repurposing) workshop
 - Posterior Predictive Simulation with P2C2M workshop
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Awards, Fellowships, and Grants

- 2023 **SSB Travel Award**, Society of Systematic Biologists.
- 2022 **Distinguished Graduate Fellowship**, Southeastern Louisiana University.
- 2022 **Graduate Research Fellowship**, Louisiana Biomedical Research Network.
- 2021 **David Ware Outstanding International Student Award**, Southeastern Louisiana University.
- 2021 **Outstanding Senior Award**, Southeastern Louisiana University.
- 2021 **Dr. Marvin L. Yates Award of Excellence**, Southeastern Louisiana University.
- 2019 **Sophomore Honors Distinction**, Southeastern Louisiana University.

Professional Memberships

- Society for the Study of Evolution (SSE)
 - Society of Systematic Biologists (SSB)
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Service and Outreach

Project Contributor

Himalayan Pariwar e.V.

Empowering Accessible Education in Nepal.

Munich, Germany

2024

President

Nepalese Student Association at Southeastern

Hammond, LA, USA

2018–2022

Planning Committee

Louisiana Sci-Comm Summit

Baton Rouge, LA, USA

2021

Vice President

Phi Kappa Phi Honor Society

Hammond, LA, USA

2020–2021

Ambassador

Honors Program at Southeastern

Hammond, LA, USA

2019–2021

Leadership Ambassador

Division of Student Affairs at Southeastern

Hammond, LA, USA

2019–2020

Senator

Student Government Association at Southeastern

Hammond, LA, USA

2019

Volunteer Judge

Southeastern Lab School Science Fair

Hammond, LA, USA

2018

REFERENCES

Prof. Dr. Sebastian Höhna

Ludwig-Maximilians-Universität München
hoehna@lmu.de

Dr. April Wright

Southeastern Louisiana University
april.wright@selu.edu